

Lecture 5, Dec 23, 2025.

Thm. The following types of functions are continuous at every number in their domains.

- Polynomials ($\text{Dom} = \mathbb{R}$).
- Rational functions ($\text{Dom}(\frac{f}{g}) = (\text{Dom } f \cap \text{Dom } g) \setminus \{g=0\}$).
- root functions $\begin{cases} \sqrt[n]{f(x)} \begin{cases} \nearrow n \text{ is even } \text{Dom } \sqrt[n]{f} = \text{Dom } f \cap \{f(x) \geq 0\} \\ \searrow n \text{ is odd } \text{Dom } \sqrt[n]{f} = \text{Dom } f. \end{cases} \end{cases}$
- trigonometric functions, $\text{Dom } \sin x = \mathbb{R}$, $\text{Dom } \cos x = \mathbb{R}$.

Example. On what intervals is each function continuous?

a). $f(x) = x^{100} - 2x^{37} + 75$.

Sln. f is a polynomial, so it's continuous on $(-\infty, \infty)$.

b). $g(x) = \frac{x^2 + 2x + 17}{x^2 - 1}$

Sln. Notice that $\text{Dom } g = \mathbb{R} - \{1, -1\}$. then since g is a fraction g is continuous on $(-\infty, -1) \cup (-1, 1) \cup (1, \infty)$.

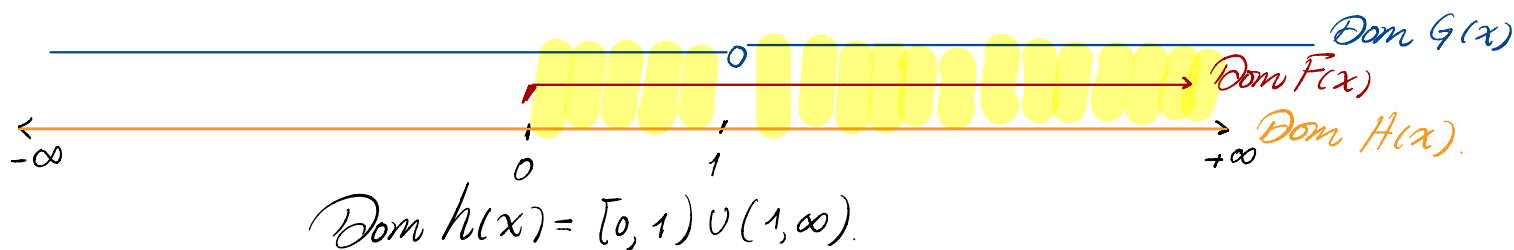
c). $h(x) = \sqrt{x} + \frac{x+1}{x-1} - \frac{x+1}{x^2+1}$

Sln. We can write $h(x) = F(x) + G(x) - H(x)$ where

$$F(x) = \sqrt{x}, \quad G(x) = \frac{x+1}{x-1}, \quad H(x) = \frac{x+1}{x^2+1}$$

$$\text{Dom } F(x) = [0, \infty), \quad \text{Dom } G(x) = \mathbb{R} - \{1\} = (-\infty, 1) \cup (1, \infty).$$

$$\text{Dom } H(x) = (-\infty, \infty)$$



Example. Evaluate $\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x}$.

Sln. By the thm $y = \sin x$ is continuous on $\mathbb{R}$ and $y = 2 + \cos x$ is also continuous on $\mathbb{R}$. Now $\frac{\sin x}{2 + \cos x}$ is continuous for x s.t $2 + \cos x \neq 0$ but notice that

$$\cos x \geq -1. \quad \text{then}$$

$$2 + \cos x \geq -1 + 2 = 1$$

So, $2 + \cos x \neq 0$ for all $x \in \mathbb{R}$. then,

$$f(x) = \frac{\sin x}{2 + \cos x} \text{ is continuous on } \mathbb{R}.$$

in particular $f(x)$ is continuous at $x = \pi$.

$$\lim_{x \rightarrow \pi} \frac{\sin x}{2 + \cos x} = f(\pi) = \frac{\sin \pi}{2 + \cos \pi} = 0.$$

Thm. If f is a continuous at b and $\lim_{x \rightarrow a} g(x) = b$ then

$$\lim_{x \rightarrow a} f(g(x)) = f(b).$$

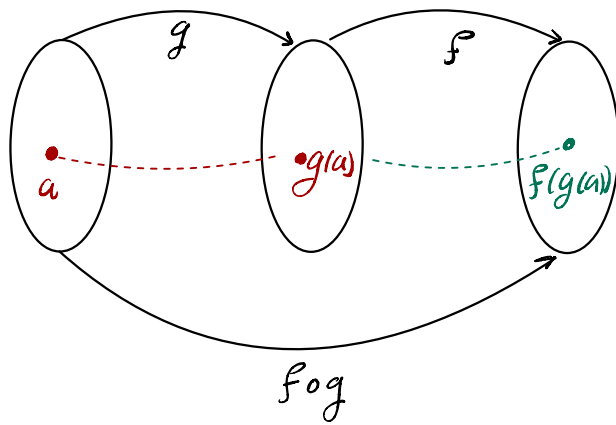
in other words

$$\lim_{x \rightarrow a} f(g(x)) = f(\lim_{x \rightarrow a} g(x)).$$

For example, $f(g(x)) = \sqrt[n]{g(x)}$ and

$$\begin{aligned} \lim_{x \rightarrow a} f(g(x)) &= \lim_{x \rightarrow a} \sqrt[n]{g(x)} \\ &= \sqrt[n]{\lim_{x \rightarrow a} g(x)}. \end{aligned}$$

Thm. If g is continuous function at a and f is continuous at $g(a)$ then the composite function $f \circ g$ given by $(f \circ g)(x) = f(g(x))$ is continuous at a .



Example. where are the following functions continuous?

a). $h(x) = \sin(x^2)$

b). $F(x) = \frac{1}{\sqrt{x^2+7} - 4}$

Sln. a). Notice that

$$h(x) = f(g(x)) \quad \text{where}$$

$$f(x) = \sin x, \quad g(x) = x^2.$$

$$(f(g(x)) = f(x^2) = \sin x^2 = h(x)).$$

Now, g is continuous on $\mathbb{R}$ (it's a polynomial) f is also continuous everywhere. Then by thm $h = f \circ g$ is continuous everywhere.

b). $F(x) = f(g(x))$ where $f(x) = \frac{1}{\sqrt{x} - 4}$ $g(x) = x^2 + 7$.

Notice that g is a continuous function on $\mathbb{R}$ b/c it's a polynomial. For $f(x)$ since it's a fraction solving $\sqrt{x} - 4 = 0$.

$$\sqrt{x} = 4$$

$$(\sqrt{x})^2 = 4^2$$

$$x = 16.$$

then, $\text{Dom } F = \mathbb{R} - \{16\}$. For $F(x)$ we need compute.

$$\sqrt{x^2+7} - 4 = 0$$

$$\sqrt{x^2+7} = 4$$

$$(\sqrt{x^2+7})^2 = 16$$

$$x^2 + 7 = 16$$

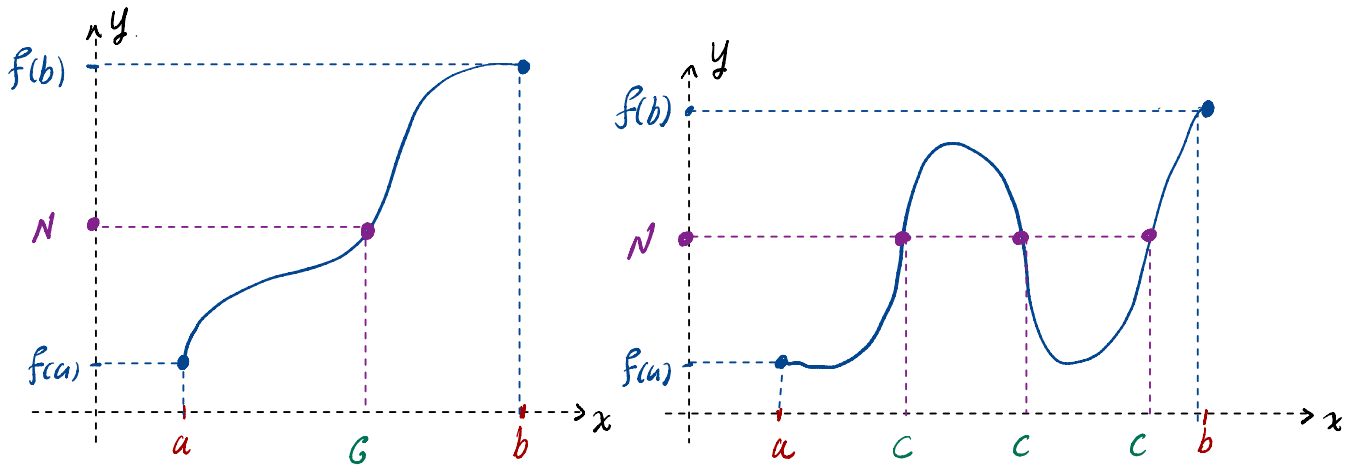
$$x^2 = 9$$

$$x = \pm 3$$

$$\text{Dom } F(x) = (-\infty, -3) \cup (-3, 3) \cup (3, \infty)$$

The intermediate Value Theorem.

Thm. Suppose that f is continuous on the closed interval $[a, b]$ and N be a number between $f(a)$ and $f(b)$, where $f(a) \neq f(b)$. Then, there exists a number $c \in (a, b)$ such that $f(c) = N$.



Example. Show that there is a solution of the equation.

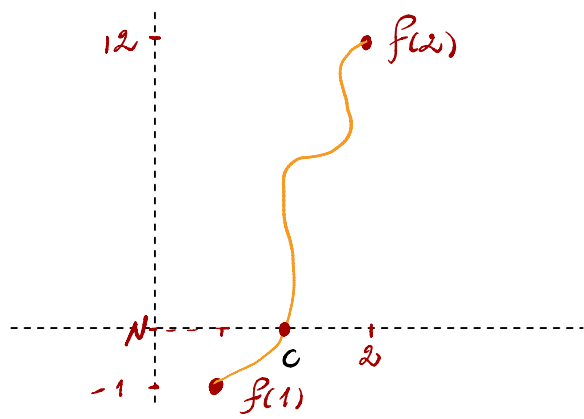
$$4x^3 - 6x^2 + 3x - 2 = 0 \quad (1)$$

between 1 and 2.

Sln: Let $f(x) = 4x^3 - 6x^2 + 3x - 2$ (1) iff $f(x) = 0$

Notice that $f(x)$ is continuous on $[1, 2]$. (b/c f is a polynomial then is continuous on $\mathbb{R}$ in particular it's continuous on $[1, 2]$)

$$\text{then, } f(1) = 4 - 6 + 3 - 2 = -1 < 0 \quad f(2) = 4(8) - 6(4) + 6 - 2 = 12 > 0.$$



So if $N=0$, $f(1) < 0 = N < f(2)$

it's a number between $f(1)$ and $f(2)$

then since f is continuous by intermediate Value theorem there is a number c between $(1, 2)$ s.t. $f(c) = 0$

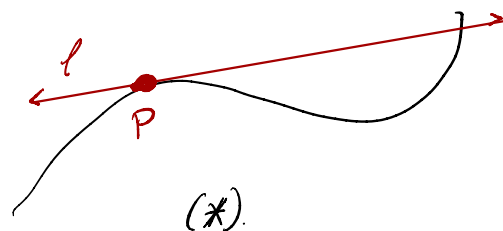
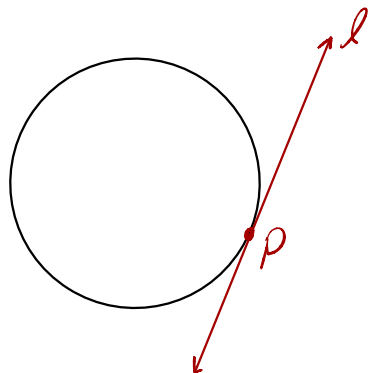
that

$$4c^3 - 6c^2 + 3c - 2 = 0$$

which is the solution for the equation (11) between 1 and 2.

The tangent and Velocity Problems.

The tangent problem. The word tangent means "touching". We can think of a tangent to a curve as a line that touches the curve and follows the same direction as the curve at the point of contact.



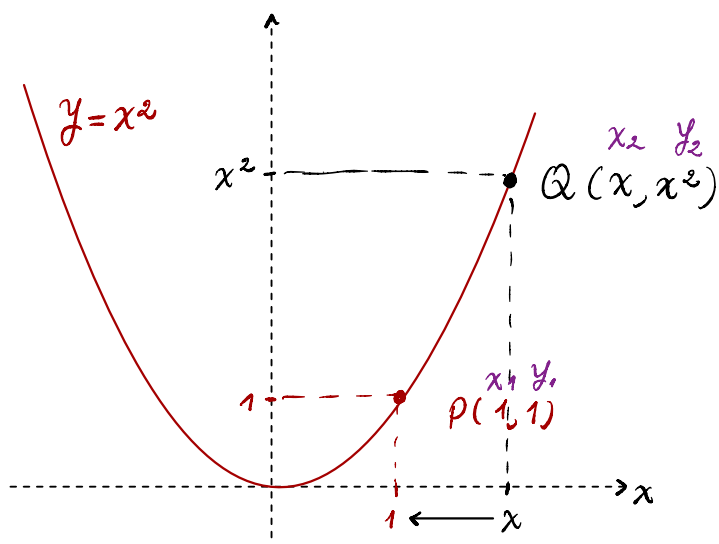
- For the circle we can say that the tangent line is a line l s.t. intersects the circle once and only once.
- For more general curves this definition is not correct in (*) l is tangent at P but intersects the curve.

Example. Find an equation of the tangent line to the curve $y = x^3$ at the point $(1, 1)$.

Sln. For a line we require two elements.

- i). slope and a point
- ii). two points.

For compute the slope we need at least two points. Here we have a problem b/c we have just one point.



then we can compute the slope of the line $\overleftrightarrow{PQ}$.

$$m_{\overleftrightarrow{PQ}} = \frac{y_2 - y_1}{x_2 - x_1} = \frac{x^2 - 1}{x - 1} \quad \begin{matrix} P \neq Q \\ x \neq 1 \end{matrix}$$

for example: $x=2$.

$$m_{\overleftrightarrow{PQ}} = \frac{4-1}{2-1} = 3. \quad \text{but the}$$

line is not tangent at P.

If we move the point Q to P then we have a tangent line.

If $Q \rightarrow P$ then that means $x \rightarrow 1$. So, we say that the slope of the tangent line is the limit of the slope of the secant lines.

$$\lim_{Q \rightarrow P} m_{\overleftrightarrow{PQ}} = m \quad \text{and} \quad \lim_{x \rightarrow 1} \frac{x^2 - 1}{x - 1} = 2.$$

Now, using the point-slope form of the equation m -slope
P - (x_1, y_1)

$$y - y_1 = m(x - x_1)$$

Here, $x_1=1$, $y_1=1$ and $m=2$.

$$y - 1 = 2(x - 1)$$

$$y = 2x - 2 + 1$$

$$y = 2x - 1 \quad (y = mx + b).$$

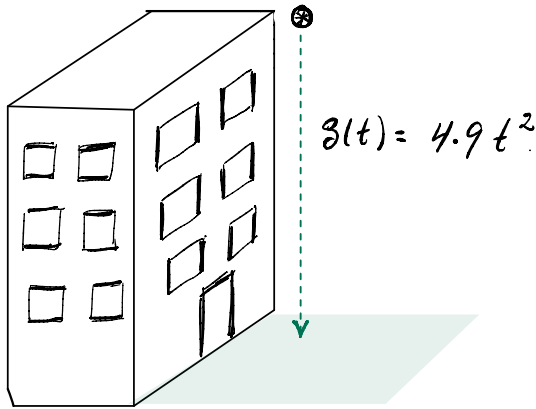
The Velocity Problem. Find the instantaneous velocity of an object moving along a straight path at a specific time if the position of the object at any time is known. In the next example, we investigate the velocity of a falling ball.

In the distance fallen after t seconds is denoted by $s(t)$ and measured in meters then the Galileo's observation says that

$$s(t) = 4.9t^2.$$

Example. Suppose that a ball is dropped from the upper observation deck of the CN Tower in Toronto, 450 m above the ground. Find the velocity of the ball after 5 seconds.

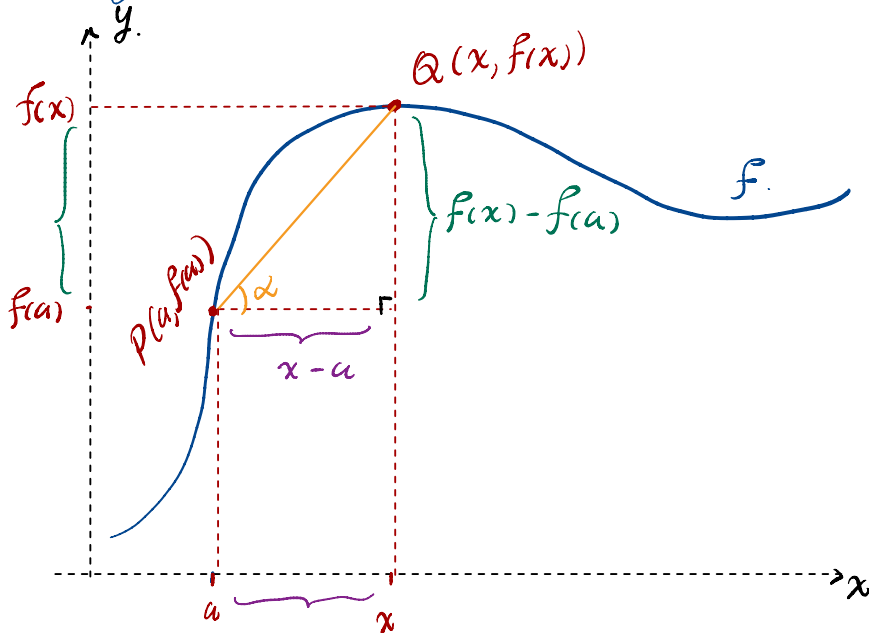
Sln. The problem here is finding the instantaneous velocity at $t=5$.



$$\text{average velocity} = \frac{\text{change in position}}{\text{change in time.}}$$

Derivatives and Rates of change.

* tangents



$$m_{PQ} = \tan \alpha = \frac{f(x) - f(a)}{x - a}$$

Def. The tangent line to the curve $y = f(x)$ at the point $P(a, f(a))$

is the line through P with slope

$$m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}.$$

Provided that this limit exists.

$$v|_{t=5} = \lim_{t \rightarrow 5} \text{average velocity} = \lim_{t \rightarrow 5} \frac{s(t) - s(5)}{t - 5}$$

recall that $s(t) = 4.9t^2$ then $s(5) = 4.9(5)^2$

$$\begin{aligned} v|_{t=5} &= \lim_{t \rightarrow 5} \frac{4.9t^2 - 4.9(5)^2}{t - 5} \\ &= \lim_{t \rightarrow 5} \frac{4.9(t^2 - 5^2)}{t - 5} \\ &= 4.9 \lim_{t \rightarrow 5} \frac{(t-5)(t+5)}{t-5} \\ &= 4.9 \lim_{t \rightarrow 5} (t+5) = 4.9(10) = 49 \text{ m/s} \end{aligned}$$

Remark. we know that $m = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$; put $h = x - a$
 $x = h + a$

$$m = \lim_{h \rightarrow 0} \frac{f(h+a) - f(a)}{h}$$

Example Find an equation of the tangent line the hyperbola $y = \frac{3}{x}$ at the point $(3, 1)$
 $a \quad f(a)$

Soln:

$$\begin{aligned} m &= \lim_{h \rightarrow 0} \frac{f(h+3) - f(3)}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3}{h+3} - \frac{1}{1}}{h} \\ &= \lim_{h \rightarrow 0} \frac{\frac{3 - (h+3)}{h+3}}{\frac{h}{1}} \\ &= \lim_{h \rightarrow 0} \frac{3 - h - 3}{h(h+3)} \end{aligned}$$

$$= \lim_{h \rightarrow 0} \frac{-\cancel{h}}{\cancel{h}(h+3)}$$

$$= - \lim_{h \rightarrow 0} \frac{1}{h+3} = -\frac{1}{3}.$$

x_1, y_1
 $P(3, 1)$

then $y - y_1 = m(x - x_1)$

$$y - 1 = -\frac{1}{3}(x - 3)$$

$$y = -\frac{1}{3}x + 1 + 1$$

$$y = -\frac{1}{3}x + 2$$

Derivatives

Def. The derivative of a function f at a number a denoted by $f'(a)$ is

$$f'(a) = \lim_{h \rightarrow 0} \frac{f(h+a) - f(a)}{h}$$

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

if this limit exists.

Example. Use the definition of derivative to find $f'(2)$ if

$$f(x) = x^2 - 8x + 9.$$

Sln. $f'(2) = \lim_{h \rightarrow 0} \frac{f(h+2) - f(2)}{h}$

$$f(h+2) = (h+2)^2 - 8(h+2) + 9 = h^2 + 4h + 4 - 8h - 16 + 9 = h^2 - 4h - 3.$$

$$f(2) = 2^2 - 8(2) + 9 = 4 - 16 + 9 = -3.$$

$$f'(2) = \lim_{h \rightarrow 0} \frac{[h^2 - 4h - 3] - [-3]}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h^2 - 4h - 3 + 3}{h}$$

$$\begin{aligned}
&= \lim_{h \rightarrow 0} \frac{h^2 - 4h}{h} \\
&= \lim_{h \rightarrow 0} \frac{\cancel{h}(h-4)}{\cancel{h}} \\
&= \lim_{h \rightarrow 0} (h-4) = -4.
\end{aligned}$$

for you! compute $f'(a)$. and $f'(5)$, $f'(10)$.